

1. K-means Clustering (K=2, Euclidean distance)

Steps to Solve:

1. Initialize two centroids randomly from the data points.
2. Assign each point to the nearest centroid using Euclidean distance.
3. Calculate the new centroids by averaging points in each cluster.
4. Repeat until assignments no longer change or meet a stopping condition.

Required Output:

A step-by-step calculation showing cluster assignments and centroid updates for each iteration

We'll perform K-means with K=2 on the given dataset and use Euclidean distance.

Steps:

1. Choose two initial centroids randomly, e.g., Sample 1 (1,5) and Sample (5,2).
2. Iter 1
 - a. Calculate Euclidean distance between each point and centroids.
 - b. Assign each point to the nearest centroid

Sample	Centroid 1 (1,5)	Centroid 2 (5,2)	Cluster
1	0.0	5.0	1
2	3.2	3.5	1
3	1.8	2.8	1
4	5.0	0.0	2
5	2.5	1.8	2
6	3.6	3.2	2
7	2.6	2.5	2

- i.
 - c. Update centroids
 - i. New Centroid 1: Mean of Cluster 1 points (1.67,3.67)
 - ii. New Centroid 2: Mean of Cluster 2 points (4.125,3.625)
3. Iter 2
 - a. Recalculate distances and reassign clusters.
 - b. Repeat steps until clusters stabilize (no changes in assignments).
 4. Result:
 - a. Final centroids and cluster assignments

2. K-means and Local Minima

- a. To demonstrate that K-means might converge to a local minimum, we can re-run the algorithm with different initial centroids and observe the sum of squared errors (SSE).

- b. K-means does not always find global optimum due to these reasons -
 - i. **Initial Centroid Dependence:** K-means' outcome relies on where initial centroids are placed, leading to different solutions based on initial positions.
 - ii. **Local Minima:** K-means can converge to local minima—suboptimal solutions where no further reduction in SSE is possible, but the clustering is not globally optimal.
 - iii. **Sensitivity to Initialization:** Poor initial centroids often result in higher SSE, trapping K-means in a less ideal clustering arrangement.
 - iv. **Mitigation Strategies:** Multiple runs with different initializations or using K-means++ (a method for better centroid initialization) help avoid local minima and improve clustering quality.

3. Optimal K with Silhouette Method

- a. Explanation of Index
- b. Selection of K
- c. Applying selected K-means to data for clustering
 - d. The Silhouette index is calculated to help determine the optimal number of clusters.
 - e. Explanation: The Silhouette Index (or Silhouette Score) is a measure of how similar a data point is to its assigned cluster compared to other clusters. It provides a way to assess the quality of clustering, helping in selecting the optimal number of clusters K

The Silhouette Score for a point i is calculated as:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

f.

where:

- $a(i)$: The average distance between point i and all other points in the same cluster (intra-cluster distance).
- $b(i)$: The smallest average distance between point i and points in a different cluster (nearest-cluster distance).

The Silhouette Score ranges from -1 to 1:

- g. **1:** Perfectly clustered (point is far from neighboring clusters).
- h. **0:** Points are on or very close to the decision boundary between clusters.
- i. **-1:** Misclassified (point is closer to a neighboring cluster than its own).

4. Hierarchical Clustering with Dendrogram

- a. We'll implement hierarchical clustering using various linkage and similarity methods.

- b. Linkage Methods in Hierarchical Clustering:
 - i. Single Linkage (Min): Clusters are merged based on the shortest distance between points in different clusters.
 - ii. Complete Linkage (Max): Clusters are merged based on the largest distance between points in different clusters.
 - iii. Average Linkage: Merges clusters by averaging distances between all points in each cluster.
 - iv. Ward's Linkage: Minimizes the increase in total within-cluster variance when merging clusters, focusing on compact, spherical clusters.
- c. Similarity Measures in Clustering:
 - i. Euclidean Distance: Measures straight-line (L2) distance, commonly used for continuous data.
 - ii. Manhattan Distance: Measures distance along axes (L1), useful for high-dimensional data.
 - iii. Cosine Similarity: Measures the cosine of the angle between vectors, useful for text and sparse data.
 - iv. Other Measures: Include Jaccard for binary data and correlation-based metrics for time series or similarity analysis.
- d. Dendrogram plotted in code output

The attached code provides a full implementation to solve each part of the assignment, including clustering calculations, SSE demonstrations, Silhouette analysis for optimal K, and hierarchical clustering with a dendrogram